



Spatial Optimisation of Fire Stations and Vehicle Allocation in the West Midlands

CASA Hackathon 2024

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Agenda

Intro	Overview of research • Background information on tasks we have chosen
Spatial & Temporal Trends	Spatial & Temporal Analysis • Analysis of spatial and temporal trends of fire events • Trends, anomalies and key insights on the distribution and frequency of fire incidents over space and time
Spatial Optimisation	1. Station Closures 2. Station Relocation • Sequential removal to determine which stations could be closed to minimise the impact • Optimal station for relocation a certain number of fire stations
Conclusion	Summary & Limitations • What we found, the limitations of our research and the implications



Task 0

Identification of Spatial and Temporal Patterns

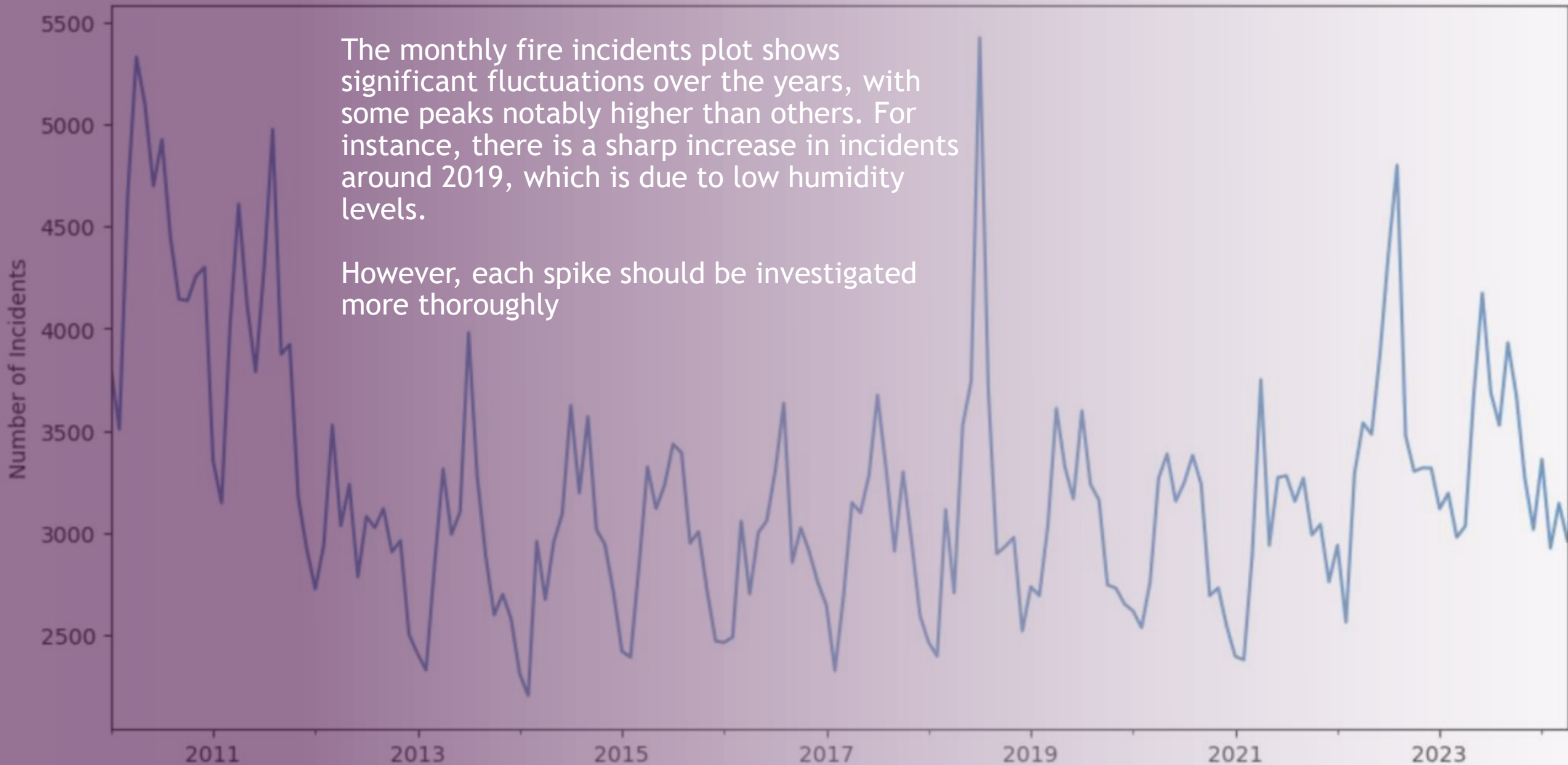
Fire station Coverage Areas

The fire station coverage area map suggests a good distribution of fire stations across the region, indicated by the different coloured clusters.

However, some areas on the outskirts have less coverage, potentially leading to longer response times in those regions.



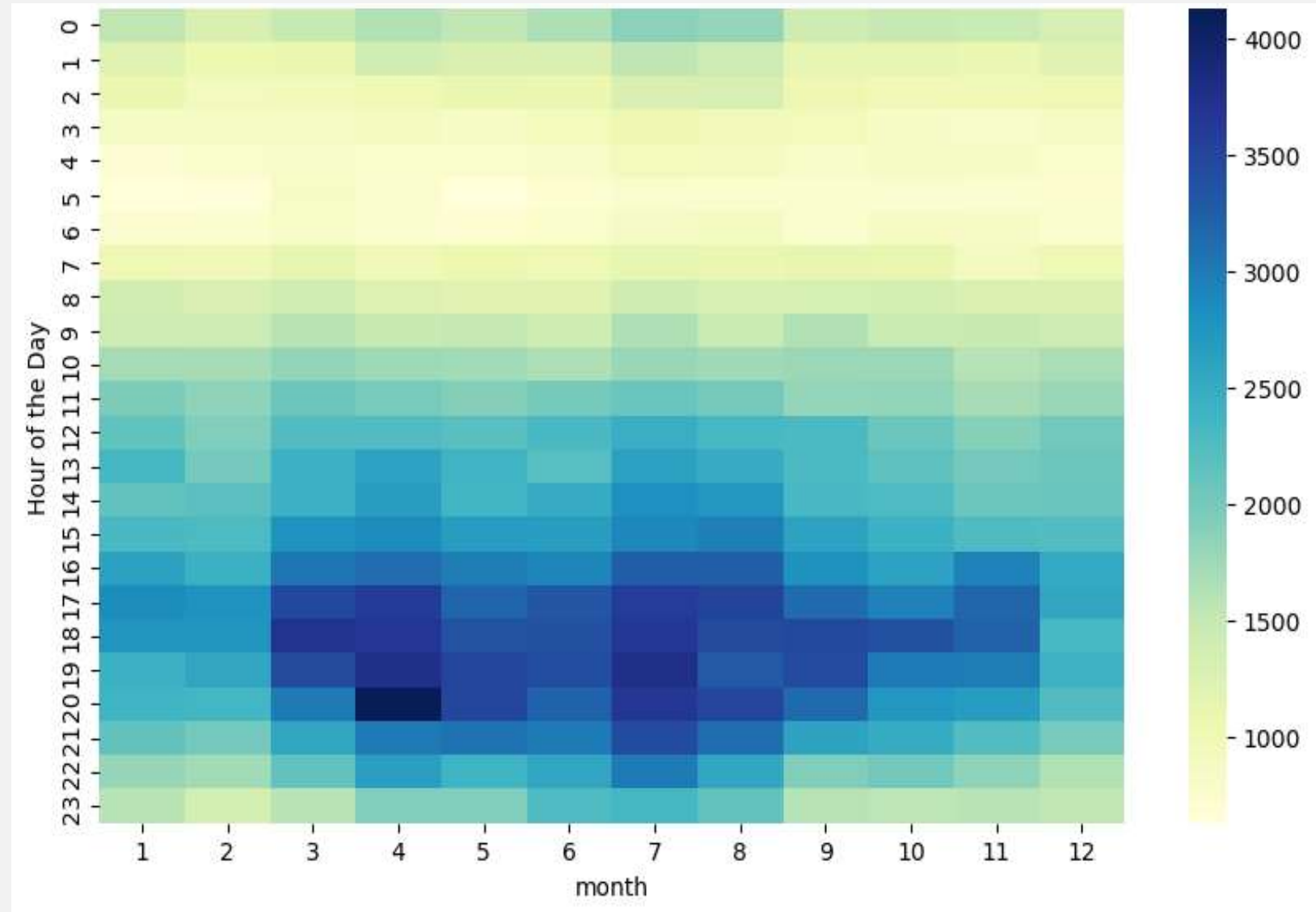
Monthly Fire Incidents 2010 - 2023



No. of Fire Incidents by Hour and Month

Higher frequency of incidents in the evening hours, particularly between 16:00 and 22:00, across most months.

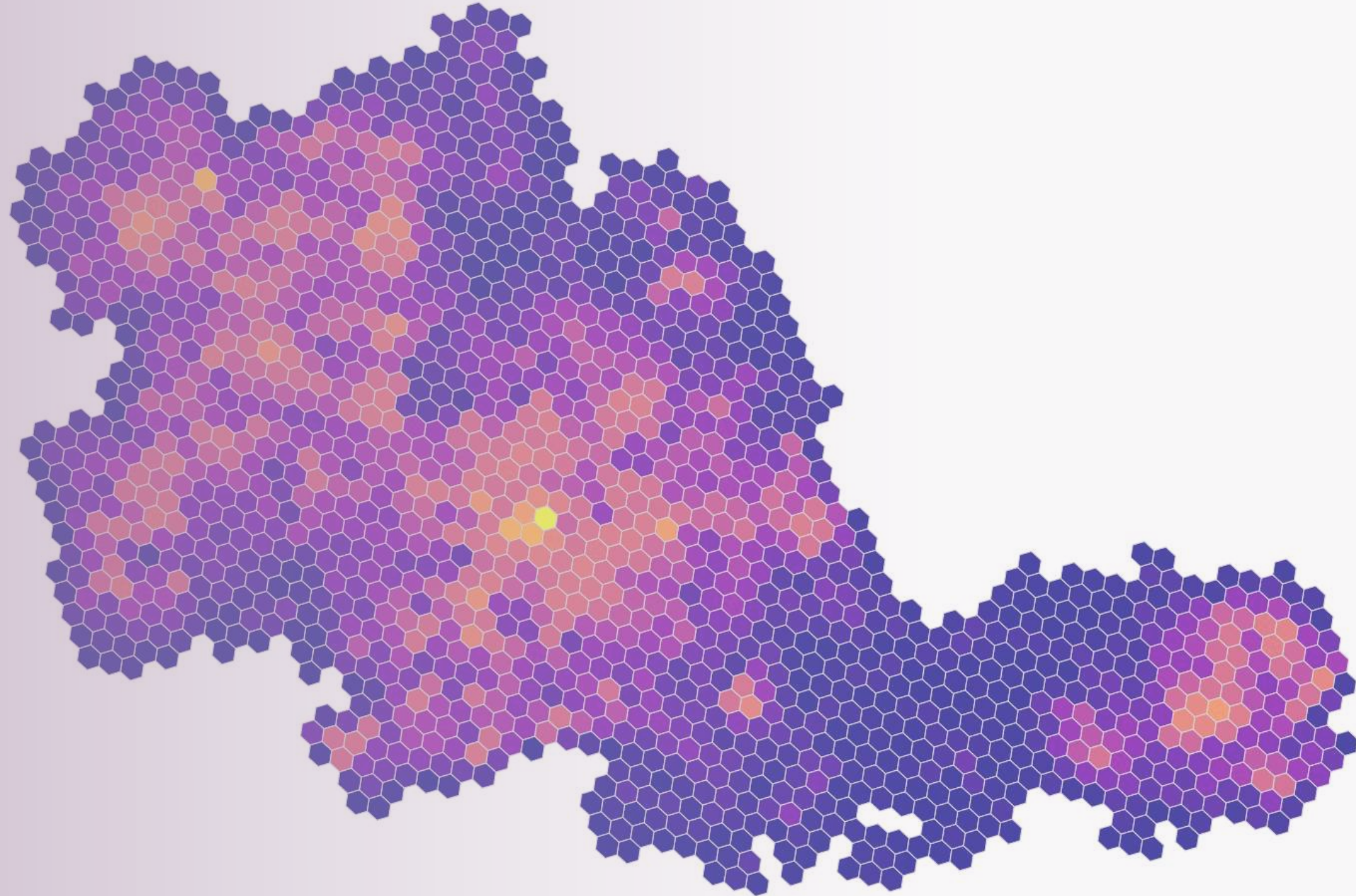
Noticeable increase in incidents during the summer months (June to August) > higher temperatures and more outdoor activities that increase fire risks.



Yearly Variation in Z-Scores 2010 - 2023

Z-Scores for Year 2010

The yearly variation in Z-scores across hexagons highlights significant hotspots where the number of incidents deviates substantially from the mean. These hotspots indicate regions with consistently higher fire risks, which could be targeted for more intensive fire prevention measures or resource allocation.



Spatiotemporal Insights

- The heatmap indicates a higher concentration of fire incidents in central areas compared to the periphery. This pattern likely represents more densely populated regions or areas with higher fire risks such as industrial zones.
- Higher frequency of incidents in the evening hours, particularly between 16:00 and 22:00, across most months.
- Additionally, there is a noticeable increase in incidents during the summer months (June to August), likely due to higher temperatures and possibly more outdoor activities that increase fire risks.



Task 2

Spatial Optimisation

Measuring the Performance of Station Allocation

Two possible patterns of quantifying the 'performance' of the network

Mitigating Impact from the Worst-Case Scenario

Average distance in the entire network calculated from the shortest path from each station to all possible points/nodes in the network

Optimising the Average Duration until Reaching Site

Total net distance travelled by fire trucks

$$D = \sum d_k n_k$$

d_k : distance to fire k from nearest fire station
 n_k : number of vehicles on site for fire k

Assumptions

- All fire vehicles arrive from the nearest station
- The distance is a perfect proxy to the time



Q1

Minimising Impact of Station Closures
due to Budget Constraints

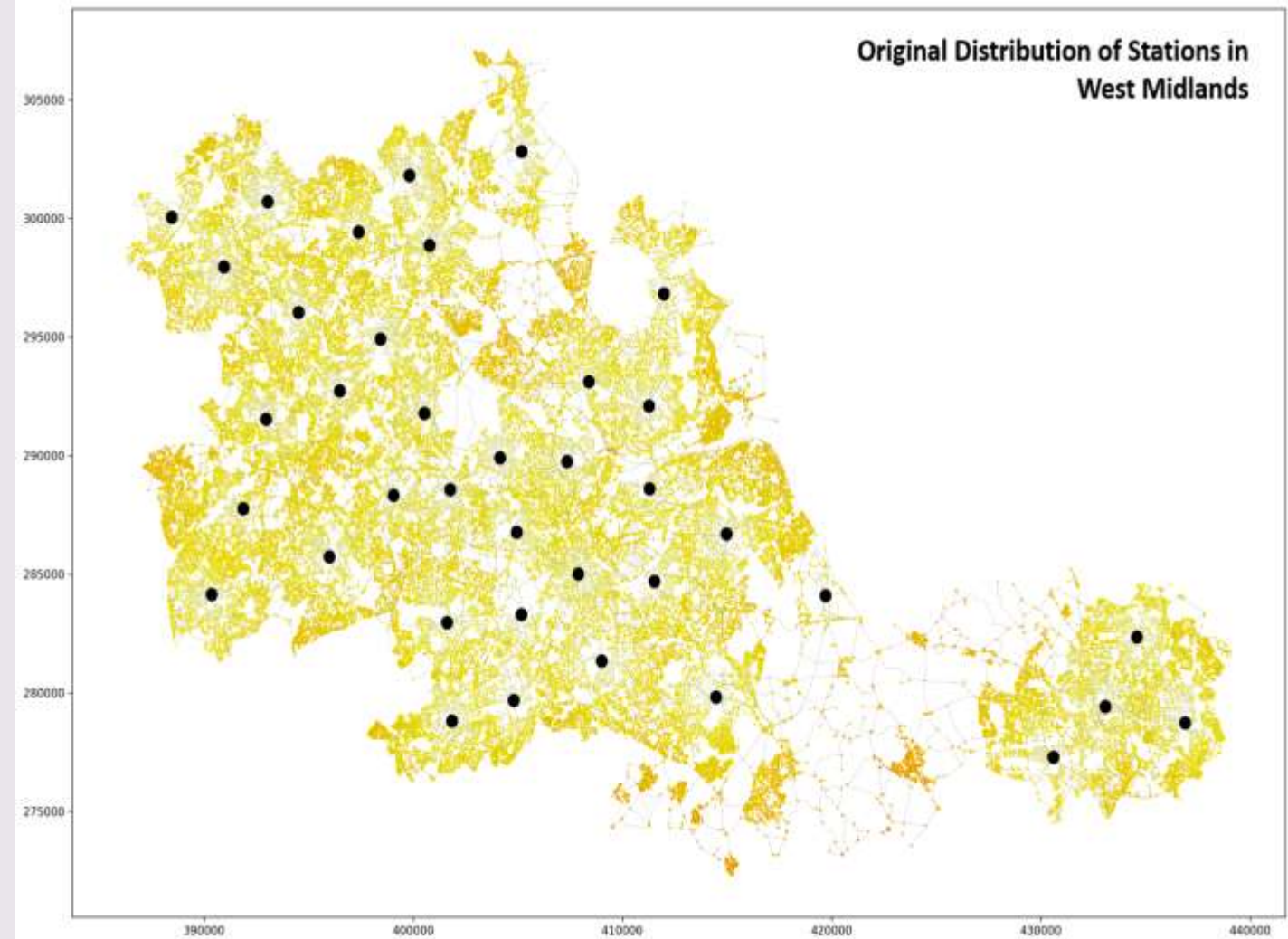
Pattern 1: Mitigating Worst Case

Removal strategy: removing the node with the lowest increase in the total average distance in the network

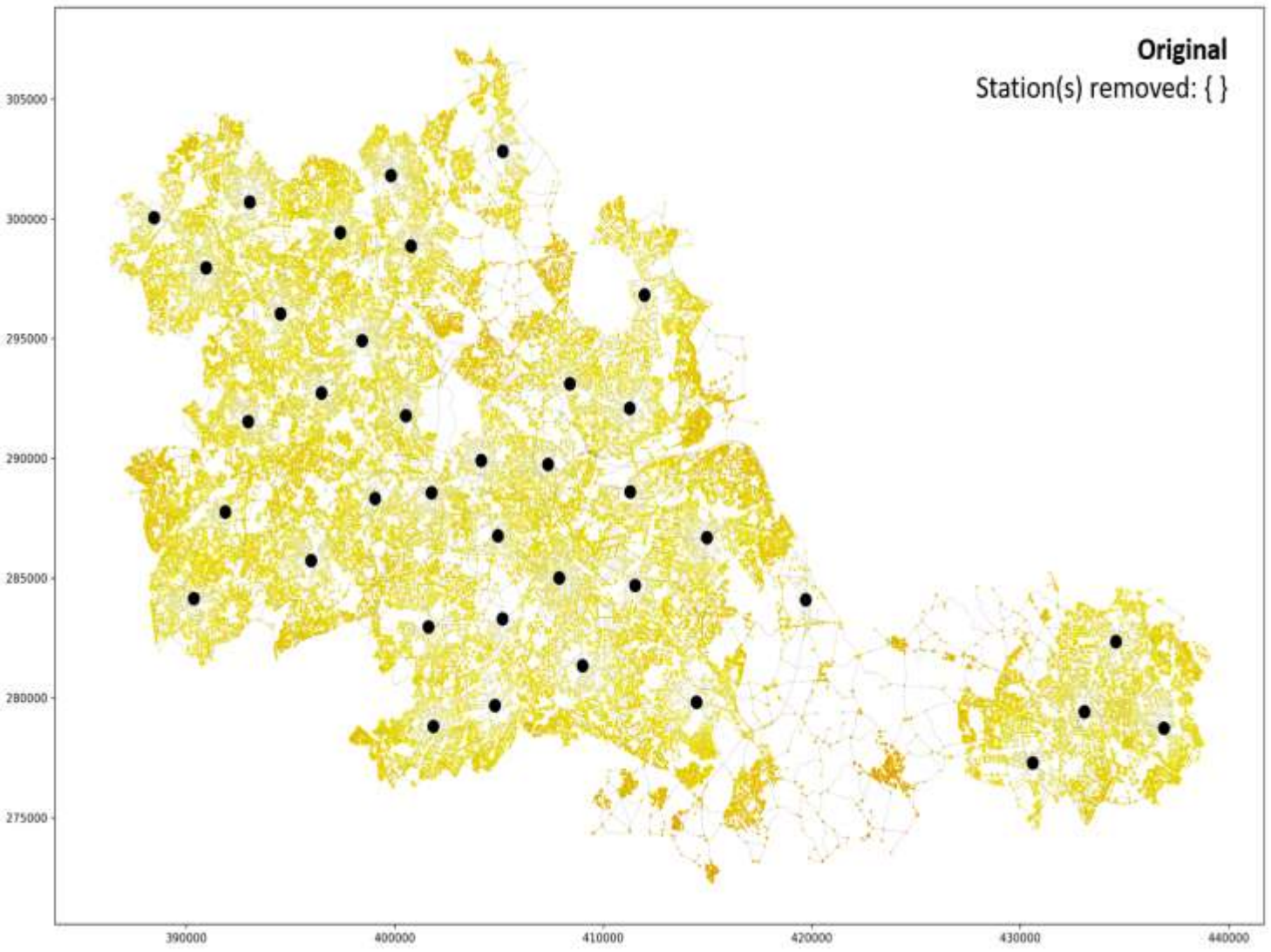
Calculate the average distance in the network when removing a single node

Remove the station with the lowest increase in average distance

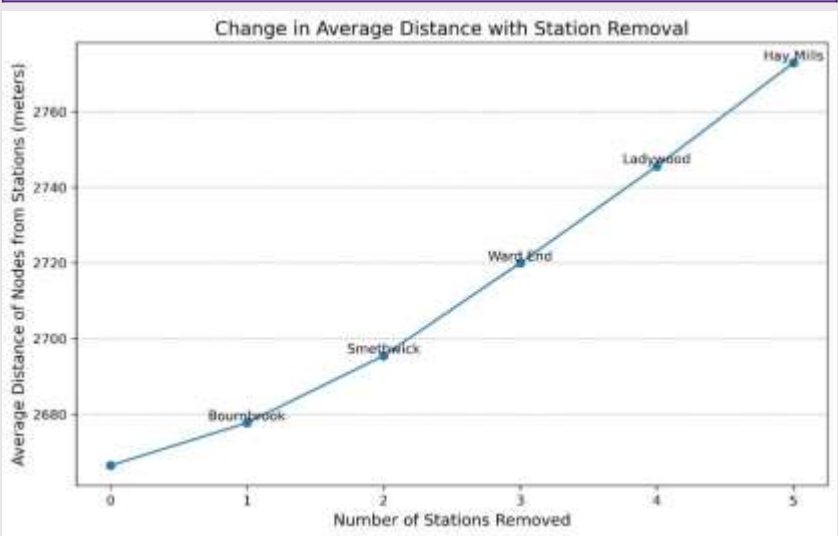
Recalculate the increase again to determine the next station to remove



Fire Stations to be Removed



Order	Station
1	Bournbrook
2	Smethwick
3	Ward End
4	Lady Wood
5	Hay Mills



Pattern 2: Optimisation of Total Cost

Removal strategy: removing the node with the lowest increase in the total net distance from the nearest fire station

Aggregate the mobilisations into H3 Grid
(Resolution 8)

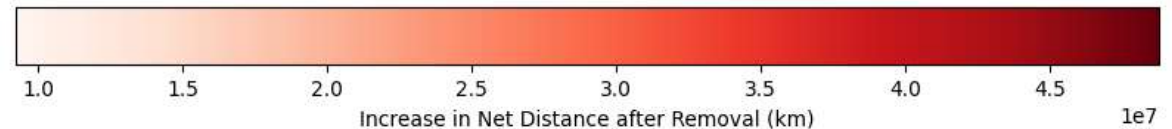
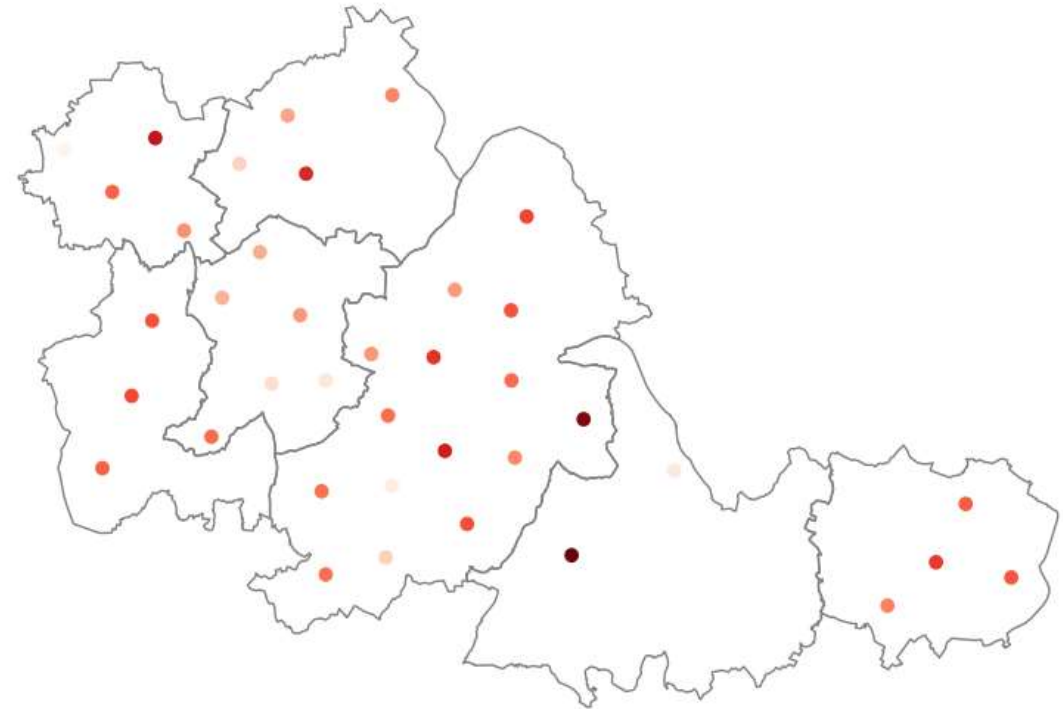


Calculate the increase in the net distance when
removing single node

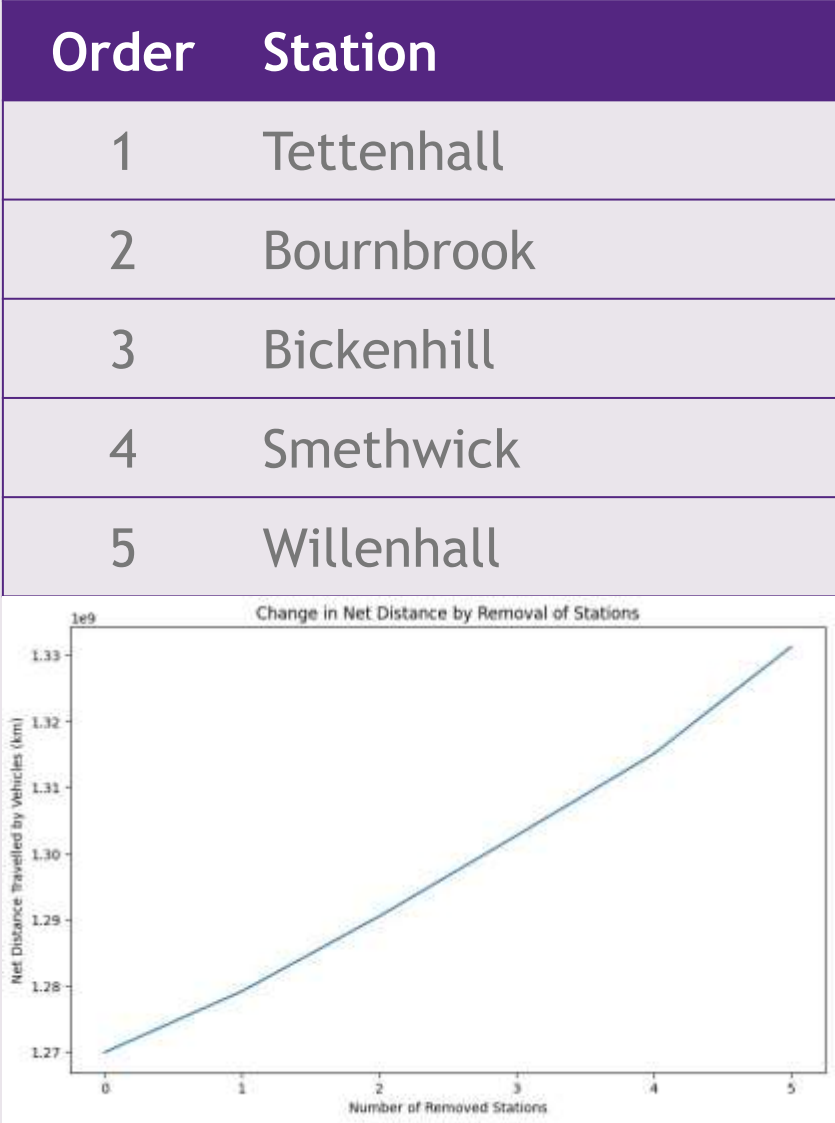
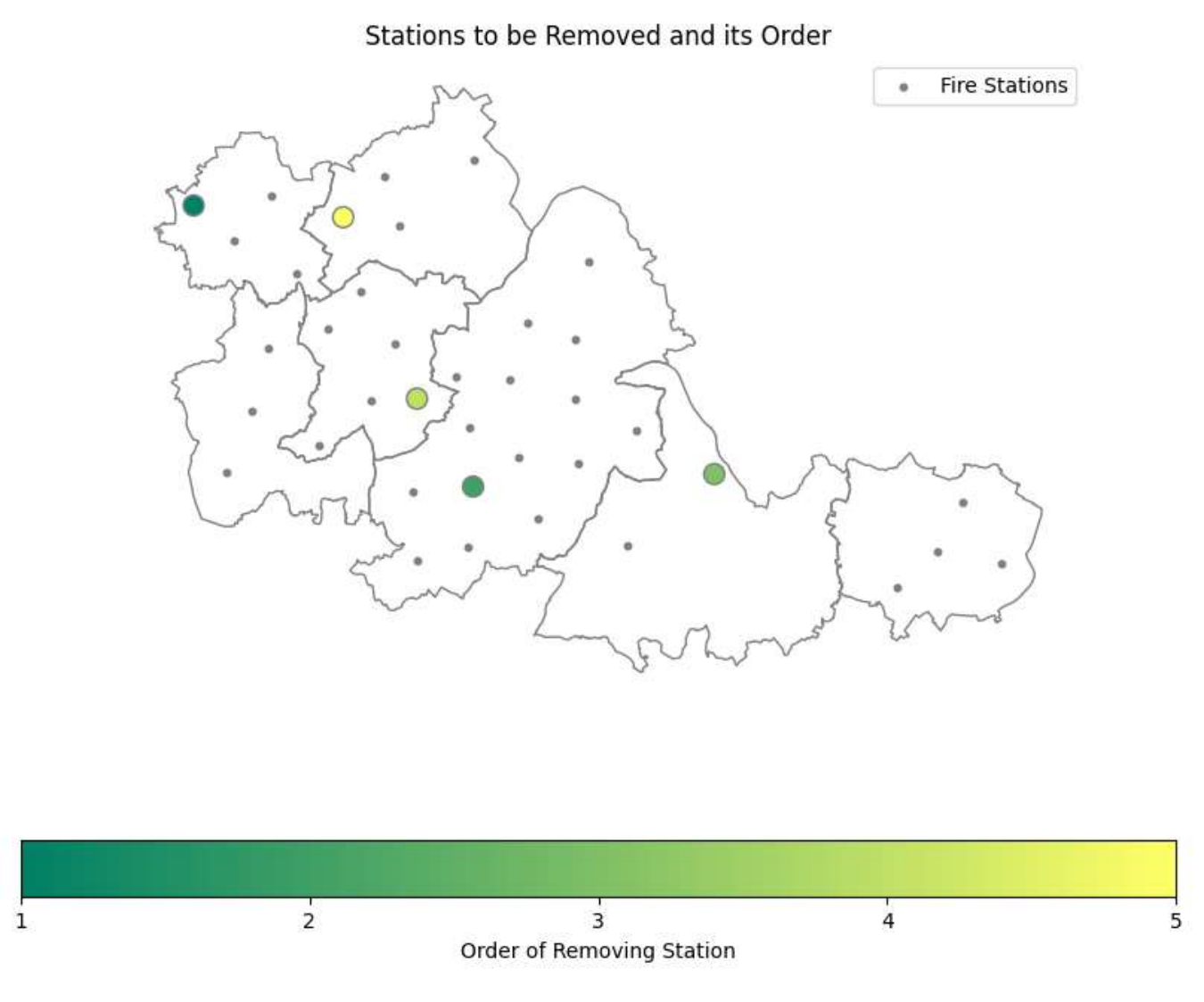
Remove the station with the lowest increase in
net distance

Recalculate the increase again to determine the
next station to remove

Increase in Net Dispatch Distance if the Fire Station was to be Removed



Fire Stations to be Removed





Q2

Identifying the Relocation of Fire
Stations to Alternative Sites

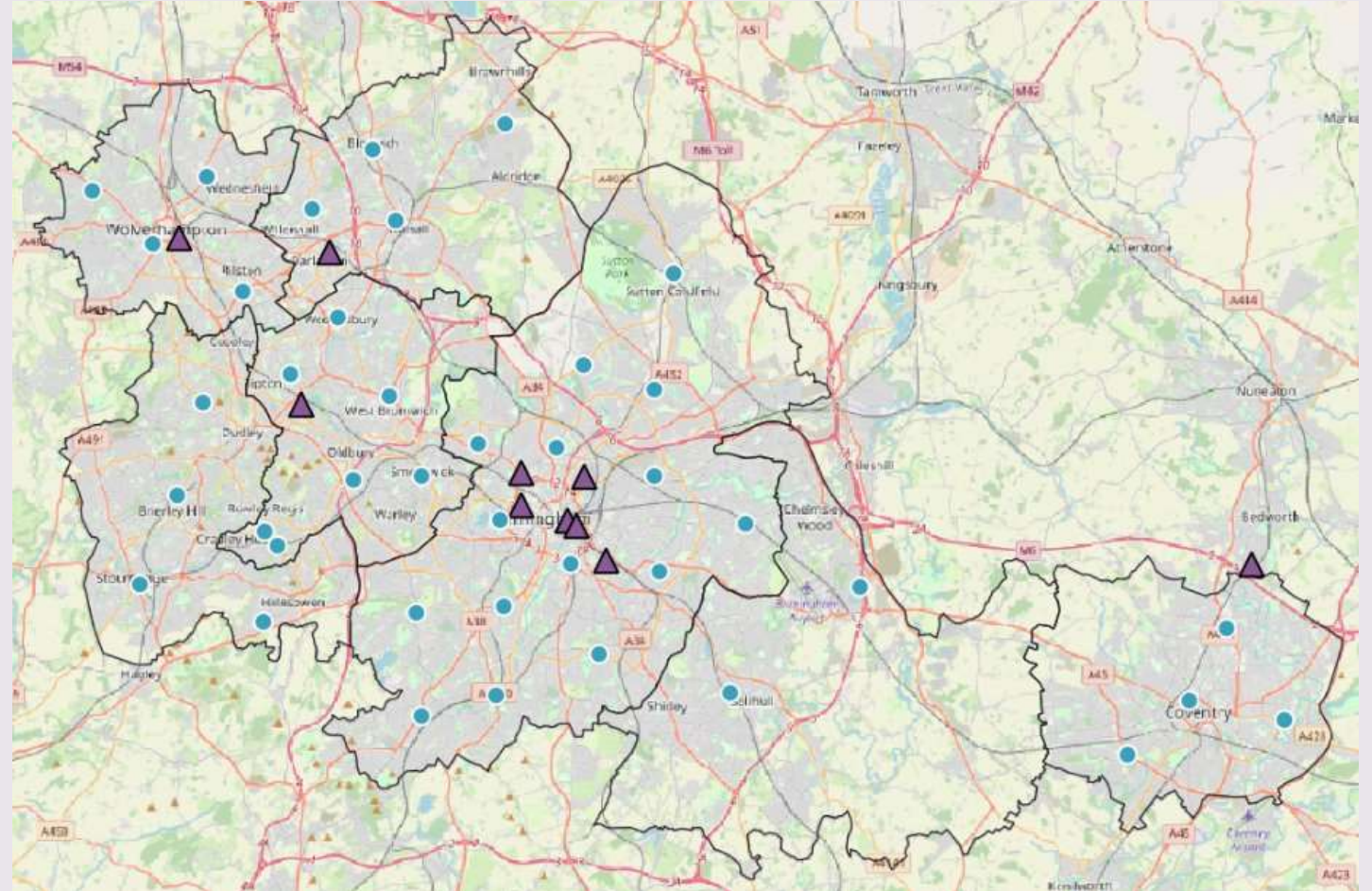
Candidates

Filter out only derelict land use

Ensure no building duplicates in dataset

Sort out top 10 areas by building area

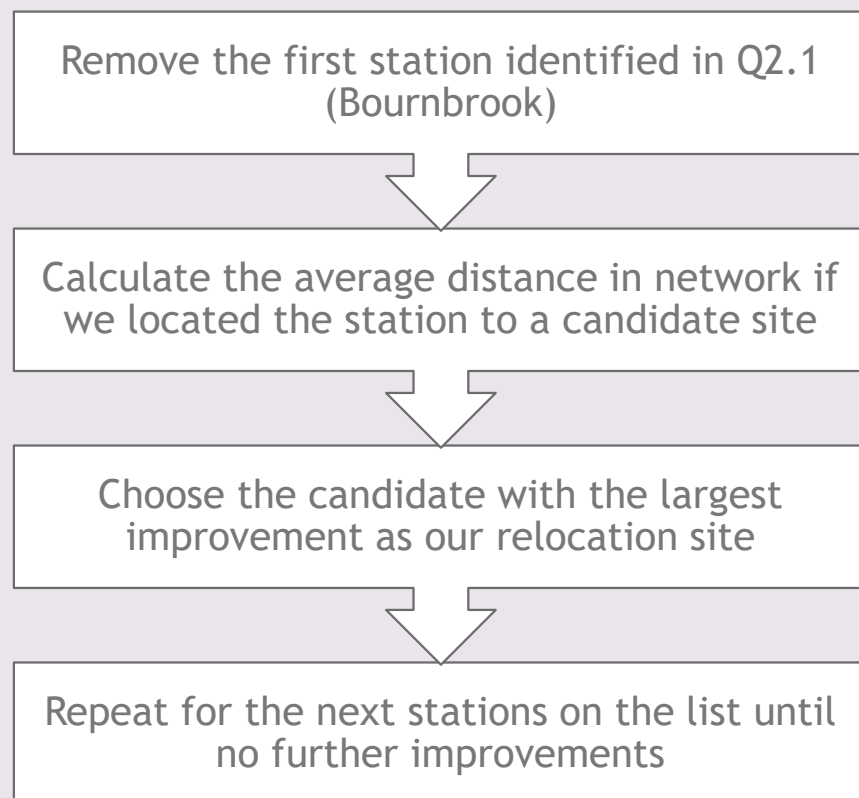
Visually inspect to ensure points have a good distribution



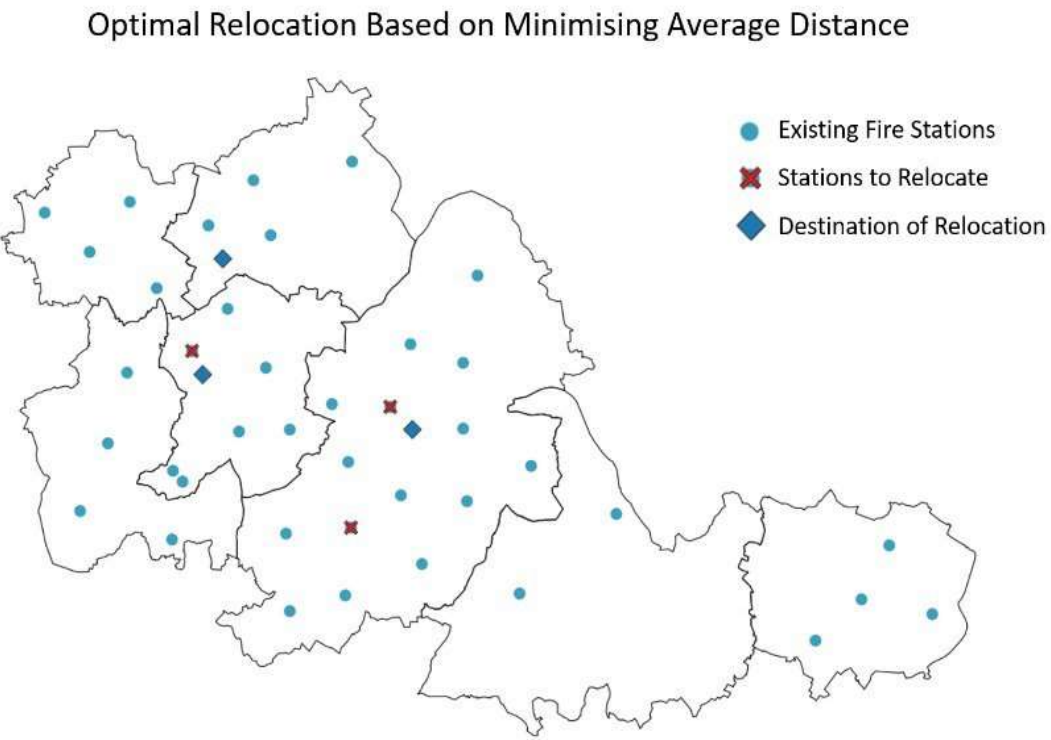
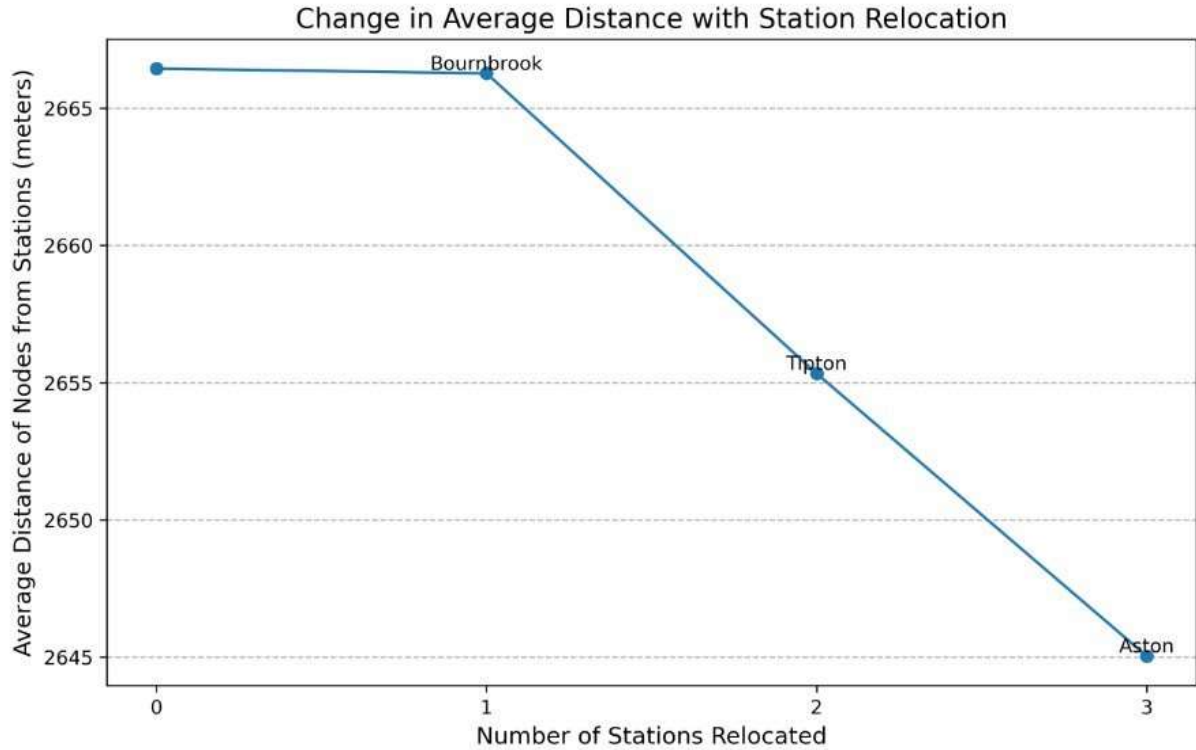
Candidate sites represented by triangles and existing stations by dots

Pattern 1: Mitigating Worst Case

Instead of entirely removing the stations identified in Q2.1, consider the relocation of these stations into the identified candidates



Optimal Relocation



The first station causes a marginal change in average distance but then drops more drastically subsequently.

Current Station	Relocation Site
Bournbrook	Lower Green, Wednesbury
Tipton	Vaughan Trading Estate, Tipton
Aston	Phoenix Business Park, Birmingham

Pattern 2: Optimisation of Total Cost

Instead of entirely removing the stations identified in Q2.1, consider the relocation of these stations into the identified candidates

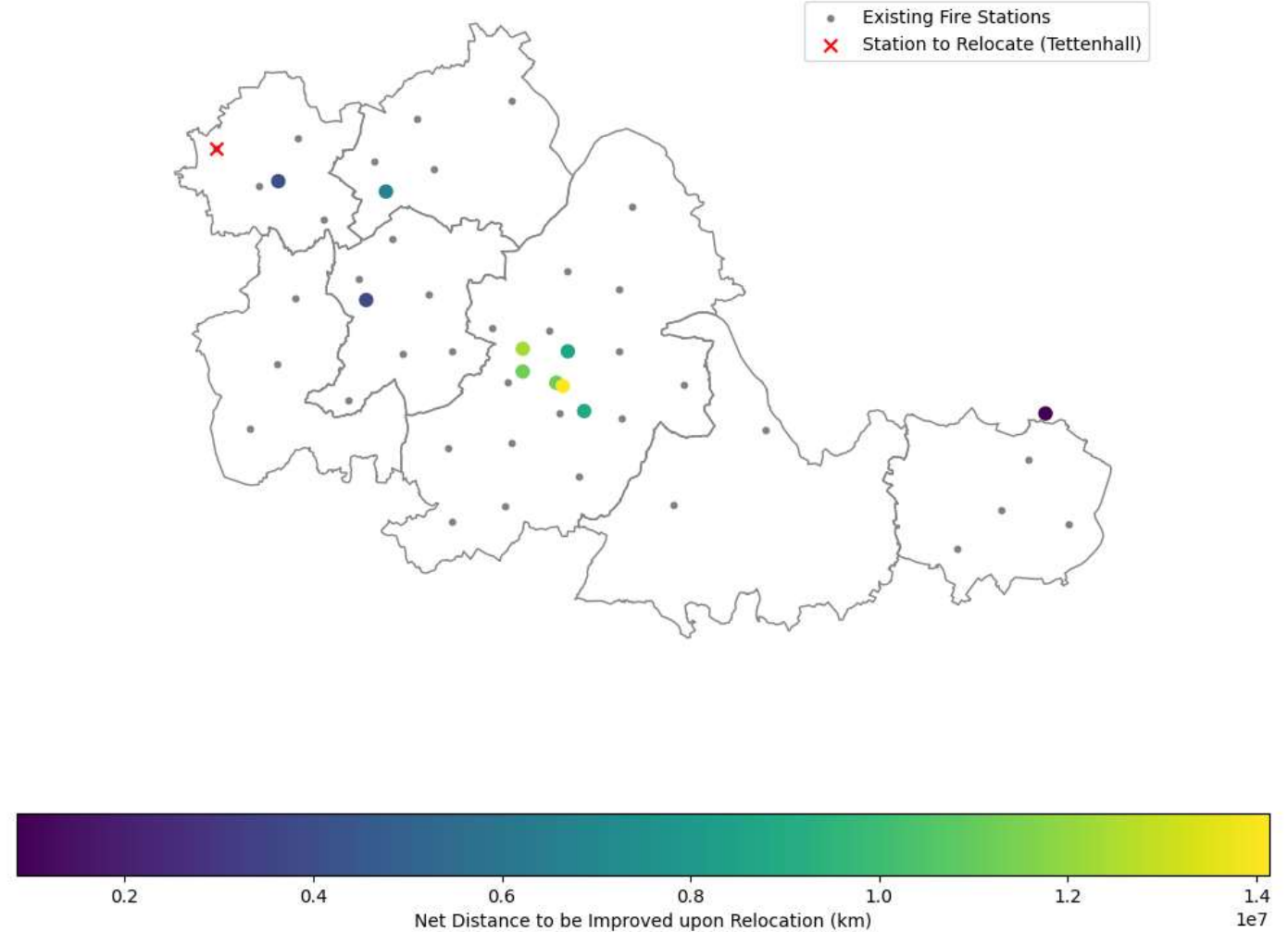
Remove the first station identified in Q2.1 (Tettenhall)

Calculate the decrease in net distance if we relocated the station to a candidate site

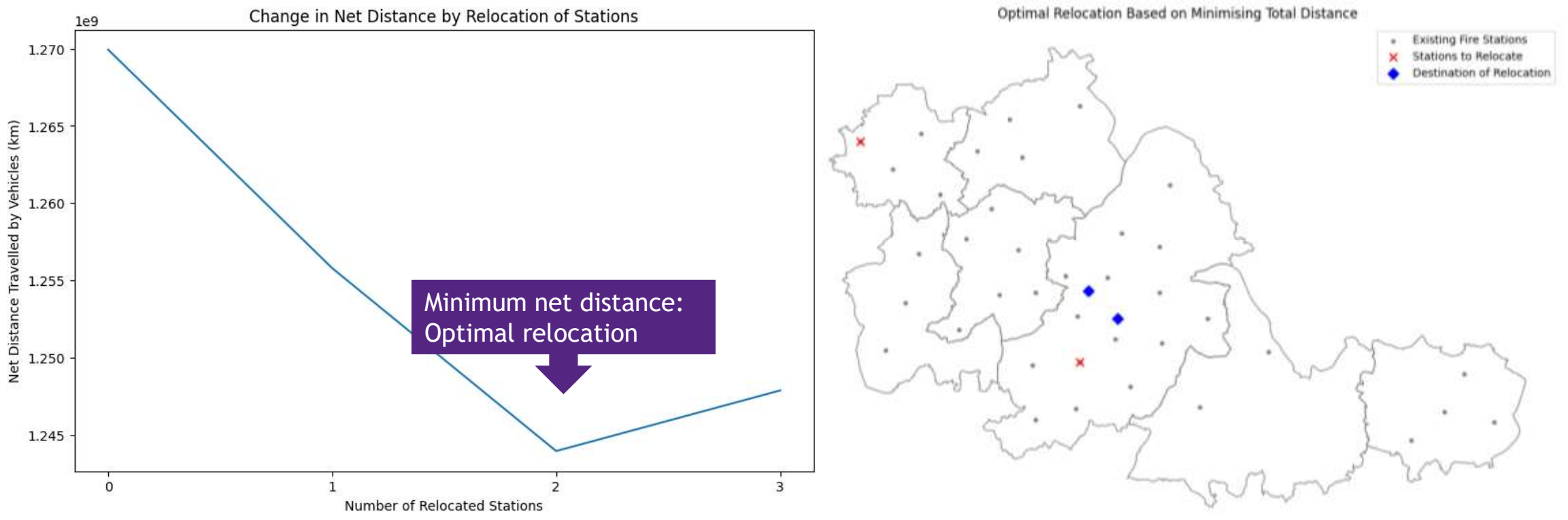
Choose the candidate with the largest improvement as our relocation site

Repeat for the next stations on the list until no further improvements

Improvement upon the Relocation of the Fire Station with the Lowest Impact



Optimal Relocation



The relocation of 2 stations minimises total travelling distance
Further relocation does not improve overall performance

Current Station	Relocation Site
Tettenhall	Heath Mill Ln, Central Birmingham
Bournbrook	Hockley Circus, Birmingham



Limitations and Future Research

Limitations

Assumptions that require further consideration:

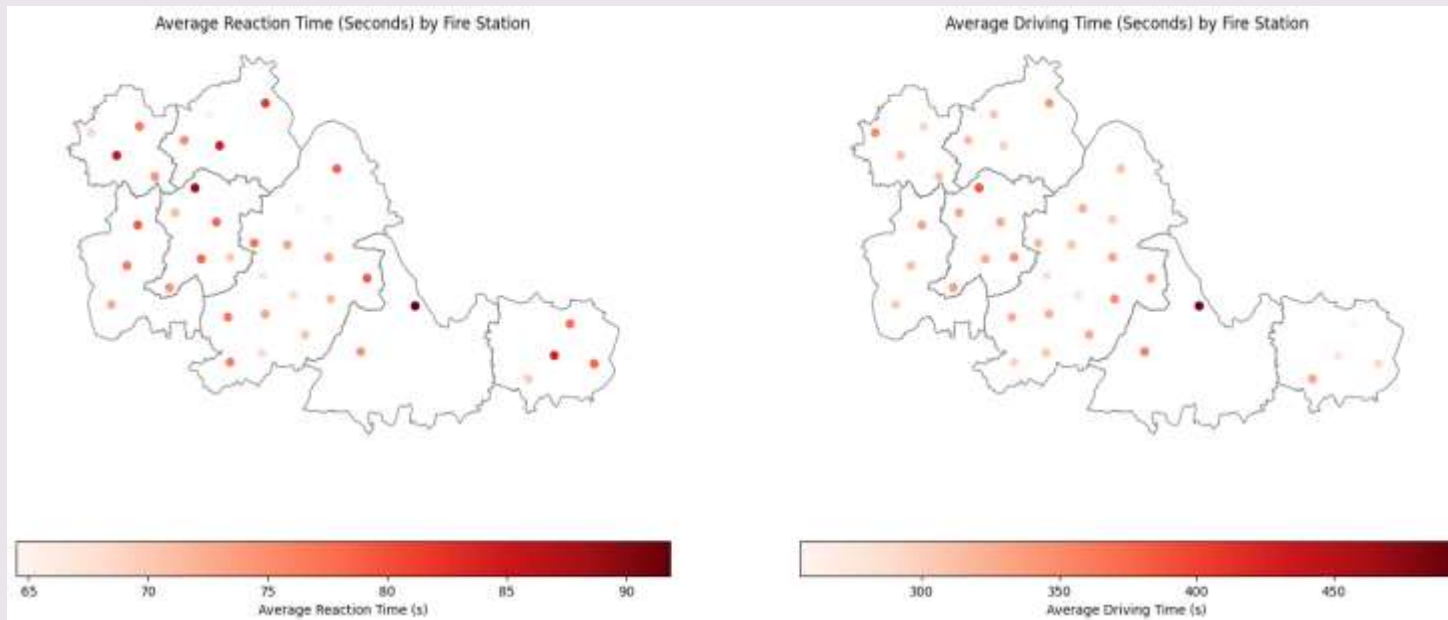
- Distance is not a perfect proxy for travel time, routing methodology has room for improvement
- Actual deployment of vehicles are not necessarily from the nearest station (e.g. vehicles may be deployed from one incident to another incident)

Future Works

Potential Methodology for Vehicle Optimisation:

Allocate the additional vehicles based on:

- Stations with high usage of current vehicles
- Stations with slower response time



End.